Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2024

Course : B.Tech | *Time : 03 Hours*
Code : 104601 | *Subject: Computer Organization and Architecture* | *Full Marks : 70*

Instructions:-
(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct answer the following (any seven question only) :*** **[2 x 7 = 14]**

(a) In a computer system, which of the following is responsible for executing instructions?
(i) ALU
(ii) RAM
(iii) CPU
(iv) Cache

(b) In a 32-bit two's complement system, what is the range of integers that can be represented?
(i) -2^{31} to $2^{31} - 1$
(ii) -2^{32} to $2^{32} - 1$
(iii) 0 to $2^{32} - 1$
(iv) -2^{31} to 2^{31}

(c) Virtual memory allows
(i) The CPU to directly access data from the hard disk
(ii) A program to use more memory than is physically available in the system
(iii) Storing data only in the RAM
(iv) Communication between different memory types

(d) In a paging system, the address generated by the CPU is known as
(i) Logical address
(ii) Physical address
(iii) Page number
(iv) Frame number

(e) Buses in computer architecture are used for
(i) Only connecting the CPU to memory
(ii) Transmitting data, addresses, and control signals between components
(iii) Powering peripheral devices
(iv) Running the operating system

(f) In a serial communication setup, data is transmitted
(i) All at once
(ii) One bit at a time
(iii) In parallel lines
(iv) Only during interrupts

(g) What is the primary purpose of pipelining in a processor?
(i) To increase the clock speed
(ii) To allow overlapping execution of instructions
(iii) To reduce the size of the instruction set
(iv) To minimize power consumption

(h) Which of the following is an example of a parallel processing system?
(i) A single-core microcontroller
(ii) A multi-core processor
(iii) A traditional desktop CPU with no threading capabilities
(iv) A simple calculator

(i) Which of the following hazards is specifically caused by the use of branches in instruction flow?
(i) Data Hazard
(ii) Structural Hazard
(iii) Control Hazard
(iv) Resource Hazard

© Apna BEU
CONFIDENTIAL DOCUMENT

Apna BEU: Syllabus, PYQs & AI
All-in-One app for Syllabus, PYQs, Result, Attendance and more.
GET IT ON Google Play

(j) What does 'throughput' refer in pipelined processors?
(i) The time taken to execute a single instruction
(ii) The total number of instructions that can be executed in a given time period
(iii) The efficiency of instruction decoding
(iv) The number of stages in the pipeline

Q.2 (a) Explain 1's and 2's complement representation. Give an example to demonstrate how it is used to represent negative numbers in binary. [7]

(b) What is the role of registers in CPU operations? Discuss the difference between general-purpose registers and special-purpose registers with examples. [7]

Q.3 (a) Consider the following page reference string: 1, 2, 3, 4, 5, 3, 4, 1, 6, 7, 8, 7, 8, 9, 7, 8, 9, 5, 4, 5, 4, 2. How many page faults would occur for the following algorithms, assuming 3 frames? [7]
(i) LRU Replacement (ii) Optimal Replacement

(b) Explain the paging technique with diagram. Also, write down the advantages and disadvantages of paging technique. [7]

Q.4 (a) A CPU needs to read data from a peripheral device at a rate of 1 KB per second. The I/O operation requires 5 microseconds to complete. If the CPU can perform operations at a rate of 1 MHz, how much CPU time is wasted in reading 1 KB of data from the peripheral device? [7]

(b) What are Buses in computer architecture? Discuss the types of buses and their functions. [7]

Q.5 (a) Define the RISC and CISC processors. Write down the differences between RISC and CISC processor. [7]

(b) Explain the concept of instruction-level parallelism (ILP) and its significance and challenges in uni-processor systems. [7]

Q.6 (a) Define the vector processing. Consider a vector processor with 8 vector registers, each capable of holding 64 elements. If a vector operation takes 10 clock cycles and the clock speed is 1 GHz, calculate the total time taken to complete 200 vector operations. [7]

(b) Discuss how pipeline depth, throughput, and latency affect overall performance metrics in pipelining. [7]

Q.7 (a) Describe the main components of a CPU and their functions. How do the Control Unit (CU) and Arithmetic Logic Unit (ALU) work together to execute instructions? [7]

(b) A cache has 64 blocks, and each block can store 32 bytes. The main memory has 4 KB of storage. Calculate [7]
i. The number of blocks in main memory. https://www.akubihar.com
ii. The number of bits required to represent the block number in main memory.
iii. The number of bits required for the cache index in a direct-mapped cache.

Q.8 (a) Discuss the concept of Asynchronous Data Communication. What are the benefits and challenges of using this approach in data transmission? [7]

(b) Briefly describe the types of hazards in pipelined processors with suitable examples. [7]

Q.9 (a) Explain and differentiate between the parallel processing and pipeline processing. [7]

(b) What is the purpose of addressing modes? Explain various addressing mode techniques. [7]

© Apna BEU
HIGHLY CONFIDENTIAL DOCUMENT EDC 145, @ 04/12/2024 10:03:55 AM

Apna BEU: Syllabus, PYQs & AI
All-in-One app for Syllabus, PYQs, Result, Attendance and more.
GET IT ON Google Play